

Функціональні залежності (FD)

tournament: id → name, start_date, prize, year_generated

team: id → name, logo, active_roster_id

player: id → nick, age

roster: id → team_id, start_date, end_date; team_id + start_date → id (якщо унікальні склади за датою)

roster_player: roster_id + player_id → (немає інших атрибутів)

match: id → tournament_id, score

match_team: match_id + team_id → (немає інших атрибутів)

team_prize: id → tournament_id, roster_id, place, money

player_prize: id → player_id, tournament_id, place, money

1NF задовільняється.

2NF задовільняється.

Транзитивна залежність: team.active_roster_id → roster.start_date/end_date порушує 3NF.

Видалення транзитивної залежності

- team.active_roster_id прибираємо з team.
- Створюємо таблицю team_active_roster:
 - team_id (FK → team.id)
 - roster_id (FK → roster.id)
 - start_date, end_date (для активності)
 - PRIMARY KEY(team_id, roster_id)

Нова схема у 3NF

CREATE TABLE

DROP TABLE IF EXISTS team_active_roster CASCADE;

DROP TABLE IF EXISTS match_team CASCADE;

DROP TABLE IF EXISTS match CASCADE;

```
DROP TABLE IF EXISTS roster_player CASCADE;
DROP TABLE IF EXISTS roster CASCADE;
DROP TABLE IF EXISTS player_prize CASCADE;
DROP TABLE IF EXISTS team_prize CASCADE;
DROP TABLE IF EXISTS player CASCADE;
DROP TABLE IF EXISTS team CASCADE;
DROP TABLE IF EXISTS tournament CASCADE;
```

```
CREATE TABLE tournament (
  id SERIAL PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  start_date DATE NOT NULL CHECK (start_date >= CURRENT_DATE),
  prize NUMERIC(12,2) NOT NULL CHECK (prize >= 0),
  year_generated INT GENERATED ALWAYS AS (EXTRACT(YEAR FROM start_date))
  STORED,
  UNIQUE(name, year_generated)
);
```

```
CREATE TABLE team (
  id SERIAL PRIMARY KEY,
  name VARCHAR(100) NOT NULL UNIQUE,
  logo BYTEA
);
```

```
CREATE TABLE player (
  id SERIAL PRIMARY KEY,
  nick VARCHAR(50) NOT NULL UNIQUE,
  age INT NOT NULL CHECK (age >= 12)
);
```

```
CREATE TABLE roster (
  id SERIAL PRIMARY KEY,
  team_id INT NOT NULL REFERENCES team(id) ON DELETE CASCADE,
  start_date DATE NOT NULL,
  end_date DATE CHECK (end_date IS NULL OR end_date > start_date),
  CONSTRAINT no_overlap CHECK (start_date < COALESCE(end_date, '9999-12-31'))
);
```

```
CREATE TABLE roster_player (
  roster_id INT NOT NULL REFERENCES roster(id) ON DELETE CASCADE,
  player_id INT NOT NULL REFERENCES player(id) ON DELETE CASCADE,
  PRIMARY KEY (roster_id, player_id)
);
```

```
CREATE TABLE team_active_roster (
  team_id INT NOT NULL REFERENCES team(id) ON DELETE CASCADE,
  roster_id INT NOT NULL REFERENCES roster(id) ON DELETE CASCADE,
  start_date DATE NOT NULL,
```

```
    end_date DATE,  
    PRIMARY KEY (team_id, roster_id)  
);
```

```
CREATE TABLE match (  
    id SERIAL PRIMARY KEY,  
    tournament_id INT NOT NULL REFERENCES tournament(id) ON DELETE CASCADE,  
    score VARCHAR(20) CHECK (score ~ '^[0-9]+:[0-9]+$' OR score IS NULL)  
);
```

```
CREATE TABLE match_team (  
    match_id INT NOT NULL REFERENCES match(id) ON DELETE CASCADE,  
    team_id INT NOT NULL REFERENCES team(id) ON DELETE CASCADE,  
    PRIMARY KEY (match_id, team_id)  
);
```

```
CREATE TABLE team_prize (  
    id SERIAL PRIMARY KEY,  
    tournament_id INT NOT NULL REFERENCES tournament(id) ON DELETE CASCADE,  
    roster_id INT NOT NULL REFERENCES roster(id) ON DELETE CASCADE,  
    place VARCHAR(20) NOT NULL,  
    money NUMERIC(12,2) NOT NULL CHECK (money >= 0)  
);
```

```
CREATE TABLE player_prize (  
    id SERIAL PRIMARY KEY,  
    player_id INT NOT NULL REFERENCES player(id) ON DELETE CASCADE,  
    tournament_id INT NOT NULL REFERENCES tournament(id) ON DELETE CASCADE,  
    place VARCHAR(20),  
    money NUMERIC(12,2) NOT NULL CHECK (money >= 0)  
);
```

team_active_roster усуває транзитивну залежність від active_roster_id.

Інші таблиці відповідають 3NF.

Всі функціональні залежності тепер чисті: всі неключові атрибути залежать лише від ключа.